📘 Title: Emerging Technologies and Trends in Information Systems

---

Slide 1: Introduction

Title: Emerging Technologies and Trends in Information Systems
Definition:
Emerging technologies are **new innovations and advancements** that are changing how people and businesses use information systems (IS).
Purpose:
To understand how modern technologies are improving efficiency, security, and decision-making in organizations.
Key Topics:

- Cloud Computing
- Internet of Things (IoT)
- Artificial Intelligence (AI) & Machine Learning (ML)
- Blockchain
- Quantum Computing

Slide 2: Cloud Computing – Definition

Definition:
Cloud computing is the **delivery of computing services** (like storage, servers, databases, and software) **over the internet**, instead of using local computers.

Example:

- **Google Drive** or **Dropbox** – store files online.
- **Microsoft Azure** or **Amazon Web Services (AWS)** – host company databases and software in the cloud.

Slide 3: Cloud Computing – Impact on IS

Impact:

- **Cost reduction:** Businesses don't need to buy expensive servers.

- **Scalability:** Resources can grow or shrink as needed.
- **Accessibility:** Employees can access data anywhere with an internet connection.
- **Backup & Recovery:** Data is automatically backed up in the cloud.

Simple View:
It's like renting computer power and storage instead of buying it.

Slide 4: Internet of Things (IoT) – Definition

Definition:
IoT means **connecting physical devices** (like cars, fridges, machines) **to the internet**, allowing them to collect and share data.

Example:

- Smart home devices (like **smart thermostats** or **security cameras**)
- **Fitbit** or **smartwatches** that monitor health
- **Smart farming systems** that monitor soil and weather

Slide 5: IoT in Business Applications

Impact on Businesses:

- **Improved efficiency:** Machines send data for predictive maintenance.
- **Real-time monitoring:** Businesses can track inventory and equipment instantly.
- **Better customer experience:** Smart devices personalize services.
- **Cost savings:** Fewer breakdowns, less waste.

Simple View:
IoT lets machines "talk" to each other and humans through the internet.

Slide 6: Artificial Intelligence (AI) & Machine Learning (ML) – Definition

AI Definition:
Artificial Intelligence is when **computers mimic human intelligence**, like thinking, reasoning, and problem-solving.

ML Definition:
Machine Learning is a **type of AI** where computers **learn from data** and improve without being directly programmed.

Examples:

- **Chatbots** that answer customer questions
- **Netflix** recommendations
- **Self-driving cars**

Slide 7: AI & ML in Decision Support

Impact on Information Systems:

- **Data analysis:** AI systems process large data sets quickly.
- **Predictions:** Helps in forecasting sales or risks.
- **Automation:** Reduces manual work by using intelligent systems.
- **Personalization:** Improves marketing strategies through data insights.

Simple View:
AI helps computers think smart, and ML helps them learn from experience.

Slide 8: Blockchain Technology – Definition

Definition:
Blockchain is a **digital ledger system** that records transactions securely across many computers so that **no one can change the records** without everyone knowing.

Example:

- **Bitcoin and cryptocurrencies**
- **Smart contracts** in businesses
- **Supply chain tracking**

Slide 9: Blockchain – Applications in IS

Impact:

- **Increased transparency:** All transactions are visible to authorized users.
- **Better security:** Data can't be easily hacked or altered.
- **Reduced fraud:** Digital records are trustworthy.
- **Faster transactions:** No middlemen needed (like banks).

Simple View:
Blockchain is like a public notebook everyone can see, but no one can erase or change secretly.

Slide 10: Quantum Computing – Definition

Definition:
Quantum computing uses **quantum bits (qubits)** instead of normal bits (0s and 1s). This allows it to perform very complex calculations much faster than normal computers.

Example:

- Used in **drug discovery**, **weather prediction**, and **cryptography**.
- Companies like **IBM** and **Google** are developing quantum computers.

Slide 11: Quantum Computing – Potential Impact on IS

Impact:

- **Faster problem solving:** Handles large datasets quickly.
- **Improved security:** Can create stronger encryption systems.
- **Better data analysis:** Helps in scientific and financial modeling.

Simple View:
Quantum computers think in many directions at once, solving problems faster than traditional computers.

Slide 12: Summary

Emerging Technologies Key Takeaways:

Technology	Main Benefit	Example
Cloud Computing	Flexible and cost-effective storage	Google Drive
IoT	Smart data from connected devices	Smart homes
AI & ML	Smarter decisions using data	Chatbots
Blockchain	Secure and transparent records	Bitcoin
Quantum Computing	Super-fast processing	IBM Quantum

Conclusion:
Emerging technologies are shaping the future of information systems, making businesses more intelligent, efficient, and secure.

12. Business Process Redesign and Automation

🔹 **Definition:**

Business Process Redesign (BPR) is the **fundamental rethinking and radical redesign** of business processes to **achieve major improvements** in performance, such as cost, quality, service, and speed.

Automation means using **technology or software** to perform business processes **with minimal human involvement.**

🔸 **1. Business Process Analysis and Redesign**

- **Definition:** The process of studying existing workflows to find inefficiencies, then **restructuring them for better performance.**
- **Steps:**
    1. Identify current processes
    2. Analyze problems (bottlenecks, delays, waste)
    3. Design improved workflows
    4. Implement and monitor
- **Example:** A bank redesigns its loan approval process by digitizing document verification, reducing approval time from 7 days to 1 day.

🔸 **2. Robotic Process Automation (RPA)**

- **Definition:** RPA uses **software robots ("bots")** to automate repetitive tasks such as data entry, report generation, and invoice processing.
- **Example:** A telecom company uses RPA to automatically update customer billing information.
- **Impact:** Reduces human error, saves time, and allows employees to focus on creative tasks.

🔸 **3. Workflow Management Systems**

- **Definition:** A software system that **manages and coordinates tasks** across departments or employees.
- **Example:** In a university, a workflow system manages student registration—automatically routing forms from students → registrar → finance → confirmation.

🔸 **4. Business Process Modeling Tools**

- **Definition:** Tools that create **visual diagrams** of business workflows using symbols (like flowcharts or BPMN).
- **Example Tools:**
  - **Bizagi Modeler**
  - **Lucidchart**
  - **Microsoft Visio**
- **Use:** Helps managers see inefficiencies and redesign processes more effectively.

🔸 **5. Continuous Process Improvement (e.g., Six Sigma)**

- **Definition:** Ongoing effort to improve processes through **measurement, analysis, and change.**
- **Six Sigma:** A data-driven approach to eliminate defects and improve quality.

- **Example:** A manufacturing company applies Six Sigma to reduce defective products from 5% to less than 1%.

💡 **Scenario:**

A hospital uses process redesign and RPA to automate patient admission. The system fills in data automatically from scanned IDs, reducing waiting times by 50%.

📊 **13. Data Governance and Quality**

🔹 **Definition:**

Data Governance ensures that an organization's data is **accurate, consistent, secure, and used responsibly**.
Data Quality Management focuses on maintaining **clean, reliable, and complete data**.

🔸 **1. Data Governance Frameworks**

- **Definition:** A structured approach that defines **who can access, manage, and use data** in an organization.
- **Example Frameworks:**
  - DAMA-DMBOK
  - COBIT
- **Benefits:** Clear rules for data ownership, security, and compliance.

🔸 **2. Master Data Management (MDM)**

- **Definition:** A process of creating a **single, consistent version** of key business data (like customer or product info).
- **Example:** A retail company ensures the same product codes are used in sales, warehouse, and accounting systems.

🔸 **3. Data Stewardship and Ownership**

- **Data Steward:** Responsible for maintaining data quality (checking accuracy and completeness).
- **Data Owner:** A person or department with legal responsibility for data.
- **Example:** In a hospital, the medical records department is the data owner; nurses act as data stewards ensuring information is correct.

🔸 **4. Data Lineage and Metadata Management**

- **Data Lineage:** Tracks where data comes from, how it moves, and where it goes (data's "life story").
- **Metadata:** "Data about data" — describes data type, size, format, and purpose.
- **Example:** In a bank, data lineage helps trace errors in financial reports to their source.

💡 **Scenario:**

A government agency uses a data governance framework to ensure that all departments record citizens' data consistently, improving accuracy in national reports.

📱 **14. Mobile and Social Media in Business**

🔹 **Definition:**

Mobile and social media technologies help businesses **connect with customers, market products, and enable mobile-based services and payments.**

🔸 **1. Mobile Application Development**

- **Definition:** The process of creating software apps for mobile devices (Android, iOS).
- **Example:** A bank develops a **mobile banking app** that lets customers check balances and transfer funds.
- **Impact:** Increases accessibility, customer satisfaction, and engagement.

🔸 **2. Social Media Integration in Business**

- **Definition:** Using social platforms (Facebook, X, Instagram, LinkedIn) to **market, engage customers, and gather insights.**
- **Applications:**
    - **Marketing:** Advertising and influencer promotions
    - **Customer engagement:** Responding to feedback in real time
    - **Analytics:** Measuring campaign performance through likes, clicks, and shares
- **Example:** Coca-Cola uses social media campaigns to promote new products globally.

🔸 **3. Mobile Security Considerations**

- **Definition:** Measures to protect mobile devices and apps from cyber threats.
- **Key Issues:**
    - Data encryption
    - User authentication
    - Malware protection
- **Example:** Banking apps use fingerprint and face ID to secure accounts.

🔸 **4. Mobile Commerce (m-Commerce) and Payments**

- **Definition:** Buying or selling goods and services through mobile devices.
- **Examples:**
    - **Apple Pay, Airtel Money, PayPal, MTN MoMo**
    - Online stores with mobile checkouts (Amazon app)
- **Impact:** Increases convenience and sales, especially in developing countries.

💡 **Scenario:**

A restaurant builds a mobile app for food ordering and payments via mobile money. Customers can order ahead, reducing queues and boosting sales.

🧱 15. Business Continuity and Disaster Recovery

🔹 **Definition:**

Business Continuity (BC) ensures that an organization's **operations continue** during or after a disaster.
Disaster Recovery (DR) focuses on **restoring IT systems and data** after disruptions.

🔸 1. Planning for Business Continuity

- **Definition:** Creating strategies to maintain essential operations during disruptions (e.g., fire, cyberattack, flood).
- **Example:** Remote work setup during power outages or pandemics.

🔸 2. Disaster Recovery Strategies for IT Systems

- **Definition:** Steps to restore IT infrastructure quickly after a failure.
- **Strategies:**
    - Data backups
    - Redundant systems
    - Cloud recovery sites
- **Example:** A company uses AWS Backup to restore databases after a ransomware attack.

🔸 3. Testing and Maintaining Continuity Plans

- **Definition:** Regular testing of BC/DR plans ensures they work effectively when needed.
- **Example:** A bank simulates server failure to check if systems switch to backup servers automatically.

🔸 **4. Cloud-Based Disaster Recovery**

- **Definition:** Using cloud services to replicate and store data in remote servers.
- **Example:** Google Cloud Backup automatically saves company files in another region.

🔸 **5. Incident Response and Crisis Management**

- **Definition:** A structured approach to **identify, contain, and recover from security incidents or disasters.**
- **Example:** An IT team isolates infected systems during a cyberattack to prevent spreading.

💡 **Scenario:**

A university hit by a cyberattack activates its continuity plan—switching to cloud backups, restoring online learning within 12 hours.

🌎 **16. International and Cross-Cultural Aspects**

🔹 **Definition:**

These aspects involve managing **global information systems** that operate across multiple countries, cultures, and legal environments.

🔸 **1. Global Information Systems and Challenges**

- **Definition:** Systems that support international operations and data sharing.
- **Challenges:**
    - Time zone differences
    - Network reliability
    - Cultural diversity

- **Example:** An international airline uses a global system for ticket reservations worldwide.

🔸 **2. Cross-Cultural Considerations in System Design**

- **Definition:** Designing systems that suit users from **different cultures and languages.**
- **Example:** A mobile app offers multiple language options and adapts to local payment methods.

🔸 **3. International Legal and Compliance Issues**

- **Definition:** Each country has different laws on data protection, copyright, and business operations.
- **Example:** The **EU's GDPR** law restricts how personal data can be collected and stored.

🔸 **4. Global Project Management and Cultural Differences**

- **Definition:** Managing international IT projects by considering **different work styles, communication, and time zones.**
- **Example:** A US team collaborates with a Japanese team; managers use flexible schedules and clear communication tools (e.g., Microsoft Teams).

🔸 **5. International Data Privacy and Cross-Border Transfers**

- **Definition:** Rules about how data moves between countries.
- **Example:** A Zambian company storing data on US-based servers must comply with both Zambian and US privacy laws.

💡 **Scenario:**

A multinational company develops a global HR system accessible in 10 countries. The system must comply with EU privacy laws, support multiple languages, and align with cultural expectations of communication.

✅ **Summary Table**

Topic	Key Idea	Example
Business Process Redesign	Streamlining work using automation	Bank automating loan approvals
Data Governance	Ensuring clean and secure data	Retail company using MDM
Mobile & Social Media	Engaging customers digitally	Company using Facebook Ads
Business Continuity	Keeping systems running after disaster	Cloud backups after cyberattack
International Aspects	Managing IS globally	Airline ticketing system